ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO1 ENGINEERING DESIGN ASSIGNMENT

Standards-Compliant Optical Fiber Link Design: From Communication Requirement to Deployment Decision

Course Outcome – CO1

Nature of Assessment: Open-Ended Engineering Design Assignment

Duration: 1 Week

Maximum Marks: 100

Primary Bloom's Levels: K3 – Apply | K4 – Analyze | K5 – Evaluate | K6 – Create/Design

THE ENGINEERING DESIGN PROBLEM

A communication service provider plans to establish a 60 km optical fiber link operating at 1550 nm to carry 10 Gb/s traffic between a central office and a business building, with provision for a later upgrade to 100 Gb/s. The link must use standard ITU-T fiber categories, tolerate tight bends at the building entry, and maintain pulse integrity, polarization performance and signal quality. The design engineer must determine whether the proposed link configuration is technically suitable before deployment.

The proposed design must satisfy the following information and constraints:

Parameter	Requirement / Given Condition
Transmission distance	60 km
Operating wavelength	1550 nm
Candidate fiber A (step-index)	n□ = 1.4500, n□ = 1.4450, core radius a = 4.0 μm
Candidate fiber B (step-index)	n□ = 1.480, n□ = 1.465, core radius a = 25 μm
Fiber losses	Absorption = 0.04 dB/km; Rayleigh scattering = 0.11 dB/km; Bending (distributed) = 0.03 dB/km
Transmitter power	+3 dBm
Connectors and splices	Four connectors, 0.4 dB each; ten splices, 0.08 dB each
Building-entry bend loss	0.8 dB with G.652.D-type fiber; 0.1 dB with G.657.A-type fiber

STAGE 1 – INTERPRET THE COMMUNICATION REQUIREMENT

1.1 Design requirements

Parameter	Given requirement
Transmission distance	60 km
Operating wavelength	1550 nm
Initial data rate	10 Gb/s
Future upgrade	100 Gb/s
Transmitter power	+3 dBm
Receiver sensitivity	−20 dBm

Required system margin	6 dB
Connectors	4 × 0.4 dB
Splices	10 × 0.08 dB
Source spectral width	0.4 nm
Input pulse width	30 ps
10 Gb/s bit period	100 ps
Required OSNR	17 dB in 0.1 nm reference bandwidth

1.2 Why is 1550 nm used?

The wavelength 1550 nm is widely used in optical communication because:

- Standard silica optical fibers have relatively low attenuation around this wavelength.
- Optical amplifiers such as erbium-doped fiber amplifiers (EDFAs) operate effectively in this wavelength region.
- It supports long-distance communication with fewer repeaters than higher-loss wavelength regions.
- Suitable single-mode fibers provide high bandwidth and low modal dispersion.

The optical signal behaves as an electromagnetic wave. Ray optics provides a useful approximation for understanding reflection and propagation, while wave optics is needed to calculate guided modes, cutoff conditions, polarization and field distribution.

1.3 Important optical-link parameters

Parameter	Purpose
Numerical aperture (NA)	Measures the light-acceptance capability of a fiber
V-number	Determines whether a fiber supports single-mode or multimode propagation
Absorption loss	Energy lost through material absorption
Rayleigh scattering	Scattering caused by microscopic refractive-index fluctuations
Bending loss	Optical power lost when the fiber bends
Chromatic dispersion	Pulse broadening caused by different wavelength components travelling at different group velocities
Modal dispersion	Pulse spreading caused by different modes travelling at different velocities
Polarization-mode dispersion (PMD)	Pulse spreading caused by different propagation delays for polarization modes
OSNR	Optical signal power relative to amplified spontaneous emission (ASE) noise
BER	Bit error rate, the fraction of received bits detected incorrectly

1.4 Relevant ITU-T fiber categories

- G.652.D: Standard single-mode fiber widely used in communication networks.
- G.655: Non-zero dispersion-shifted single-mode fiber, designed to provide controlled chromatic dispersion in its intended operating region.
- G.657.A: Bend-insensitive single-mode fiber suitable for access networks and building-entry applications.

For this link, single-mode propagation is preferred because it eliminates intermodal dispersion and supports high bandwidth over long distances.

STAGE 2 – INVESTIGATE AND SELECT THE FIBER / PROPAGATION CONFIGURATION

The numerical aperture (NA) determines the light-gathering ability of an optical fiber. It is calculated using:

$$NA = \sqrt{(n_1^2 - n_2^2)}$$

The normalized frequency, or V-number, determines whether a fiber operates in single-mode or multimode conditions:

$$V = (2\pi a/\lambda) \times NA$$

For Fiber A, with refractive indices of 1.4500 and 1.4450 and a core radius of 4 μm:

- Numerical aperture = **0.1203**
- V-number = **1.95**

Since V is less than 2.405, Fiber A supports single-mode operation at the specified wavelength.

For Fiber B, with refractive indices of 1.480 and 1.465 and a core radius of 25 μm:

- Numerical aperture = **0.2102**
- V-number = **21.28**
- Approximate number of guided modes = $V^2/2$ = **226**

Fiber B supports multiple modes and is therefore less suitable for this long-distance, high-speed link because of modal dispersion.

Fiber selection: Single-mode fiber is preferred for the main transmission span. G.655 fiber is selected to reduce chromatic dispersion, while G.657.A fiber is used at the building entrance to minimize bending losses.

STAGE 3 – DEVELOP AND EVALUATE THE OPTICAL POWER BUDGET

- Absorption loss = 0.04 dB/km
- Rayleigh scattering loss = 0.11 dB/km
- Bending loss = 0.03 dB/km

Total attenuation per kilometre:

$\alpha = 0.04 + 0.11 + 0.03$

= **0.18 dB/km**

For a 60 km link:

Fiber loss = 0.18×60

= **10.8 dB**

Connector loss:

$4 \times 0.4 =$ **1.6 dB**

Splice loss:

$10 \times 0.08 =$ **0.8 dB**

Using G.657.A entry fiber with a bend loss of 0.1 dB, the total passive link loss is:

Total loss = 10.8 + 1.6 + 0.8 + 0.1

TL = **13.3 dB**

Received power:

$P_r = 3 - 13.3 =$ **−10.3 dBm**

Power margin:

Margin = −10.3 − (−20) = **9.7 dB**

The available margin exceeds the required 6 dB by 3.7 dB. Therefore, the passive link satisfies the specified power-budget requirement under the given assumptions.

Rayleigh scattering contributes $0.11 \times 60 = 6.6$ dB over the link, making it the largest of the specified distributed-loss components.

STAGE 4 – INVESTIGATE AND SELECT THE DISPERSION / POLARIZATION-MANAGEMENT CONFIGURATION

Chromatic dispersion occurs because different wavelength components of an optical pulse travel at different speeds through the fiber. This causes pulse broadening and may lead to inter-symbol interference.

Pulse broadening is estimated using:

$\Delta\tau = |D| \times \Delta\lambda \times L$

where D is the dispersion coefficient, $\Delta\lambda$ is the source spectral width, and L is the fiber length.

For G.652.D fiber:

$\Delta\tau = 17 \times 0.4 \times 60 =$ **408 ps**

For G.655 fiber:

$\Delta\tau = 4.5 \times 0.4 \times 60$

= **108 ps**

At 10 Gb/s, the bit period is 100 ps. The calculated broadening exceeds this period for both fibers, so dispersion compensation is required under the simplified model used here.

The approximate combined pulse width for G.655, including the 30 ps input pulse, is:

$\Delta\tau(\text{total}) = \sqrt{(108^2 + 30^2)}$

= **112.1 ps**

For dispersion-compensating fiber (DCF), the required length is estimated by:

$L(\text{DCF}) = (D \times L)/|D(\text{DCF})$

Using D = 4.5 ps/(nm·km),

L = 60 km

D(DCF) = −100 ps/(nm·km):

L(DCF) = (4.5 × 60)/100 = **2.7 km**

The DCF attenuation is:

Loss = 2.7 × 0.5 = **1.35 dB**

DCF compensates for accumulated chromatic dispersion, but its attenuation must be included in the power budget. The final design should be validated using the actual transceiver specifications and complete dispersion measurements.

STAGE 5 – EVALUATE OSNR AND THE AMPLIFICATION STRATEGY

PMD occurs when different polarization components of an optical signal travel at slightly different speeds. It can distort high-speed data signals, particularly as the bit period becomes shorter.

The approximate PMD delay is:

$PMD = D_{pmD} \times \sqrt{L}$

Given a PMD coefficient of 0.1 ps/√km and a link length of 60 km:

PMD = 0.1 × √60 = **0.775 ps**

The specified PMD limit is 0.1 times the bit period.

At 10 Gb/s, the bit period is 100 ps and the limit is 10 ps. At 100 Gb/s, the bit period is 10 ps and the limit is 1 ps.

The calculated PMD of 0.775 ps satisfies both limits. However, the 100 Gb/s upgrade still requires verification of the complete transmission system.

STAGE 6 – VERIFY AND CRITICALLY EVALUATE THE COMPLETE LINK

OSNR indicates the ratio between optical signal power and amplified spontaneous emission (ASE) noise power, measured within the same reference bandwidth.

Initially, signal power is −16 dBm and ASE noise power is −39 dBm.

Initial OSNR = −16 − (−39) = **23 dB**

The amplifier provides 14 dB signal gain and increases the ASE noise level by 16 dB.

Output signal power = −16 + 14 = **−2 dBm**

Output noise power = −39 + 16 = **−23 dBm**

Output OSNR = −2 − (−23) = **21 dB**

The required OSNR is 17 dB, so the remaining OSNR margin is:

21 − 17 = **4 dB**

The amplifier meets the stated OSNR requirement under the given assumptions. However, the additional noise reduces OSNR by 2 dB, so amplifier noise performance and the complete link OSNR should be verified before deployment.

FINAL SYSTEM DESIGN

Parameter	Selected design value
Link length	60 km
Operating wavelength	1550 nm
Data rate	10 Gb/s
Main transmission fiber	G.655
Building-entry fiber	G.657.A
DCF length	2.7 km
Passive loss before DCF	13.3 dB
Total loss including DCF	14.65 dB
Received power including DCF	−11.65 dBm
Power margin including DCF	8.35 dB
Calculated PMD	0.775 ps
Output OSNR	21 dB
Required OSNR	17 dB

Engineering Decision and Conclusion

Decision: Conditional YES for 10 Gb/s operation.

The proposed optical link satisfies the specified power-margin, PMD, and OSNR requirements based on the supplied parameters and simplified calculations. G.655 fiber reduces the dispersion-compensation requirement, while G.657.A fiber minimizes bending losses at the building

entrance. A 2.7 km DCF section provides the estimated dispersion compensation, although its additional attenuation must be considered.

Before deployment, the link should undergo end-to-end testing of received power, dispersion, OSNR, connector and splice losses, and overall system performance.

For a future 100 Gb/s upgrade, the design must be re-evaluated for transceiver compatibility, modulation format, dispersion tolerance, OSNR requirements, and nonlinear effects.

In conclusion, the proposed design is a suitable starting point for a 60 km, 10 Gb/s optical communication link. Final deployment approval should be based on measured link performance and the specifications of the selected equipment.